

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Smart Study Generator Agent

Domain of Project – Education

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Problem Statement -

Problem Statement :Smart Study Generator Agent

The Challenge:Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

1. **Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.
2. **Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.
3. **Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).
4. **Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

Proposed Solution -

Proposed Solution: Agentic AI-Powered Smart Study Generator

The proposed solution is StudyGenAI, an Agentic AI-powered Smart Study Generator built using IBM Langflow, IBM Orchestrate, and IBM Granite Models.

The platform acts as an intelligent study companion that understands the student's academic requirements and autonomously performs multiple tasks such as:

- Collecting learning material from multiple sources
- Summarizing large documents
- Creating flashcards and quizzes
- Answering questions using uploaded study material
- Generating concept maps
- Preparing personalized study schedules
- Tracking learning progress

Unlike traditional AI chatbots, the system uses Agentic AI, where multiple AI agents collaborate to accomplish complex educational tasks automatically

Technology Used:

IBM Langflow- Builds AI workflows and connects multiple AI agents.

IBM Orchestrate- Manages and automates all AI tasks and agents

IBM Granite Models (watsonx.ai)- Generates summaries, quizzes, flashcards, and answers.

Python (FastAPI/Flask)- Backend APIs

React.js- Frontend Dashboard

PostgreSQL- Stores users, study plans, and quiz results

ChromaDB / Vector DB- Stores document embeddings for RAG

OCR (Tesseract/EasyOCR)- Extracts text from images and scanned notes

Speech-to-Text- Converts voice questions into text

IBM Cloud Object Storage- Stores uploaded files

Langflow component Used-

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized nutrition insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations
- 4) **Chat output** -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.
- 5) **File component** - Stores and retrieves user data, meal logs, and nutritional datasets for analysis and tracking.
- 6) Add name of any other component used.

Langflow workflow -

IBM watsonx Orchestrate

English

Contact Sales

Buy Now

29 trial days left

AL

Home / Manage agents / Smart Study generator Agent

Test agent

Deploy

Smart Study generator Agent

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Welcome to smart study Assistant!

Get Smart Study.wi to watsonx Orchestrate

Maximum 100 characters allowed in welcome message edit field.

Quick start prompts

Pre-set messages sent to the agent to start the conversation.

Add prompt +

Study Plan for the Subjects

Notes and quiz Generator of the Subject

Assignments and exam help

Agent style

Define how your agent interprets prompts and decide what action to take.

Default Recommended

ReAct

Preview

Running in debug mode

study plan for english

Smart Study generator Agent 3:25 PM | Show Reasoning

Topic: English – Parts of Speech

Summary:

Words in English are grouped by the role they play in sentences. Understanding parts of speech helps you parse meaning, build correct sentences, and improve writing and reading comprehension.

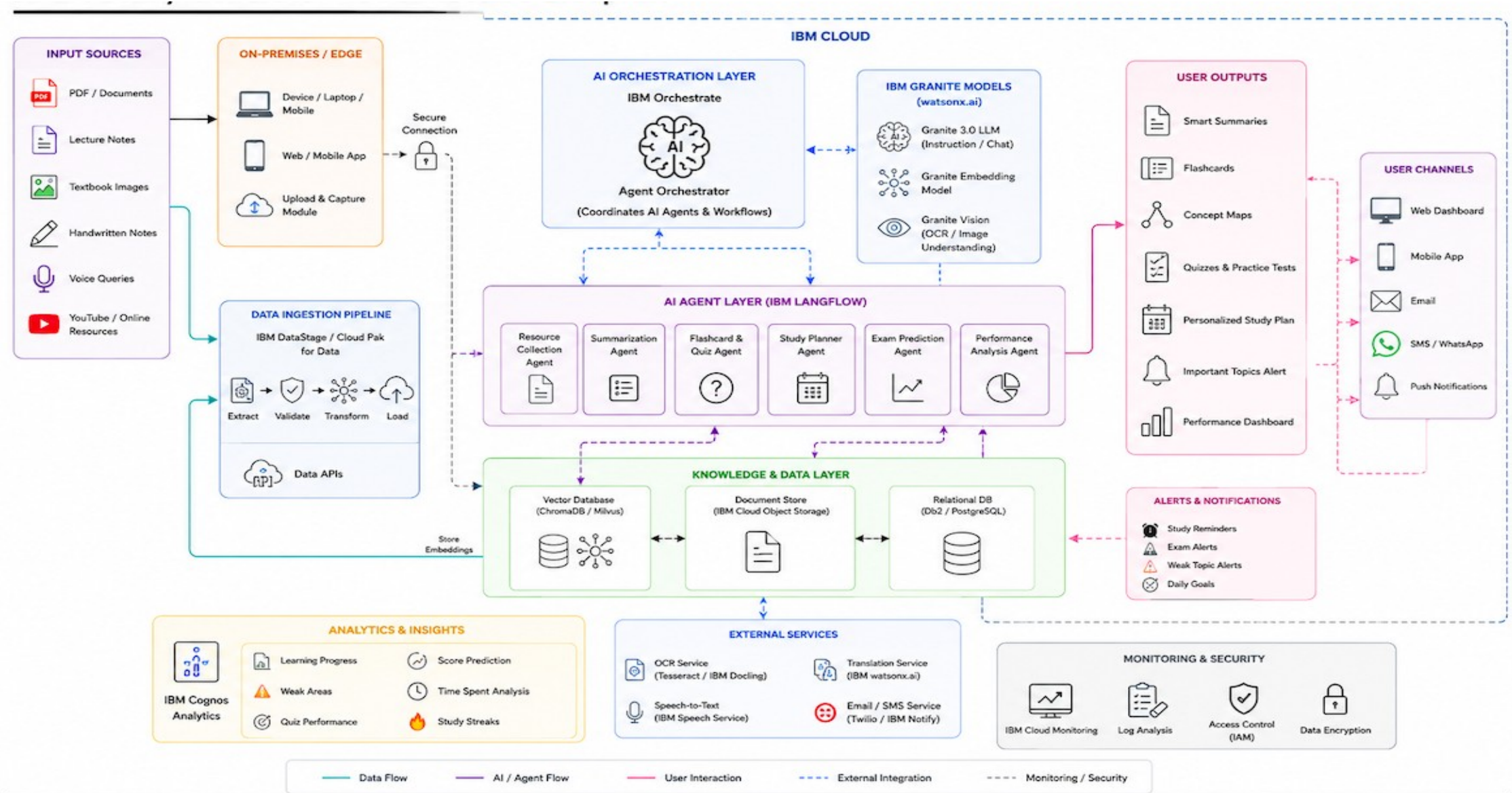
Key Points

- Five core categories: Noun, Pronoun, Verb, Adjective, Adverb (plus others like preposition, conjunction, interjection).
- Each part has a specific function (e.g., nouns name people/places, verbs show action or state).
- Many words can serve multiple roles depending on context.

Save as test

Type something...

Architecture blueprint-



Role of Agentic AI in the solution

Role of Agentic AI in Smart Study Generator

Agentic AI is the core component of the Smart Study Generator. Instead of acting like a simple chatbot,

1. Resource Collection Agent

Collects learning materials from notes, PDFs, textbooks, research papers, images, and online tutorials.

Organizes all resources into a structured knowledge base.

2. Content Processing Agent

Analyzes the uploaded content.

Generates concise summaries, key points, concept maps, and easy-to-understand explanations using IBM Granite Models.

3. Personalized Study Planner Agent

Creates a customized study schedule based on the syllabus, available study time, exam dates, and learning progress.

Recommends daily study goals and revision plans.

4. Exam Prediction Agent

Analyzes previous year question papers and syllabus.

Identifies frequently asked topics and recommends high-priority chapters for exam preparation.



Novelty and Uniqueness

Novelty

- Uses Agentic AI instead of a simple chatbot.
- Combines notes, PDFs, images, voice, and online resources into one study platform.
- Automatically generates summaries, flashcards, quizzes, and concept maps.
- Creates personalized study plans based on the student's progress.
- Predicts important exam topics using previous year question papers.

Uniqueness

- Multi-Agent AI: Different AI agents work together for summarization, quiz generation, study planning, and progress tracking.
- Multimodal Learning: Supports text, PDFs, images, and voice inputs.
- Personalized Learning: Adapts study plans and recommendations to each student.
- Progress Dashboard: Tracks learning progress, weak areas, and exam readiness.
- IBM AI Ecosystem: Built using IBM Langflow, IBM Orchestrate, and IBM Granite Models.

Git Hub Link

GitHub Repository

Repository Name: Smart Study Generator

GitHub Link:

<https://1drv.ms/u/c/A022AB0626242DC1/>

IQC_35POsefXT4niO_UOV43CAZt2NB4UEoAdCeUFntP91To

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Future Scope

- AI Tutor: Introduce a virtual AI tutor that provides real-time explanations, doubt-solving, and personalized guidance.
- Multi-Language Support: Enable learning in multiple regional and international languages to make education more accessible.
- Learning Style Adaptation: Personalize content based on the student's preferred learning style (visual, auditory, or reading/writing).
- LMS Integration: Integrate with platforms like Moodle, Google Classroom, and Microsoft Teams for seamless learning management.
- Advanced Predictive Analytics: Use AI to predict exam performance, identify learning gaps early, and provide targeted recommendations.
- AR/VR Learning: Incorporate Augmented Reality (AR) and Virtual Reality (VR) for interactive visualization of complex concepts in subjects like science and engineering..

Certificates Earned:

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Amruta Ladkat

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Completion Certificate

This certificate is presented to

Amruta Ladkat

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Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 14 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.